Ai Face Detection System

1. datasets

Stores metadata about the training dataset.

Column	Data Type	Description
dataset_id	INT (PK)	Unique dataset identifier
dataset_name	VARCHAR(100)	FER2013 Kaggle Challenge
image_format	VARCHAR(50)	48×48 grayscale
pixels_per_image	INT	2304
train_size	INT	28709
test_size	INT	3589

2. emotion_labels

Maps numeric class IDs to emotion names.

Column	Data Type	Description
emotion_id	INT (PK)	Label ID (0–6)
emotion_name	VARCHAR(30)	Emotion category

Example records:

emotion_id	emotion_name
0	Angry
1	Disgust

emotion_id	emotion_name
2	Fear
3	Happy
4	Sad
5	Surprise
6	Neutral

3. video_processing_pipeline

Stores each preprocessing stage in order.

Column	Data Type	Description
step_id	INT (PK)	Pipeline step
step_description	TEXT	Processing action

Example:

step_id	step_description
1	Analyze video frame by frame
2	Convert image to grayscale
3	Detect face using OpenCV Cascade Classifier
4	Handle multiple faces
5	Resize image to training dimensions
6	Convert image into model input
7	Predict emotion

4. cnn_layers

Describes the CNN architecture.

Column	Data Type	Description
layer_id	INT (PK)	Layer identifier
layer_type	VARCHAR(50)	Layer category
activation	VARCHAR(30)	Activation function (if applicable)
description	TEXT	Layer purpose

Example:

layer_type	activation	description
Conv2D	ReLU	Feature extraction
MaxPooling2D	NULL	Down-sampling
BatchNormalization	NULL	Normalize activations
Flatten	NULL	Flatten feature maps
Dense	Softmax	Output probabilities

5. data_augmentation

Stores image augmentation parameters.

Column	Data Type	Description
augmentation_id	INT (PK)	Record identifier
zoom_range	DECIMAL(3,2)	0.20
rotation_range	INT	10

Column	Data Type	Description
width_shift_range	DECIMAL(3,2)	0.10
height_shift_range	DECIMAL(3,2)	0.10
horizontal_flip	BOOLEAN	TRUE
vertical_flip	BOOLEAN	FALSE

SQL DDL

```
CREATE TABLE datasets (
   dataset_id INT PRIMARY KEY,
   dataset_name VARCHAR(100),
   image_format VARCHAR(50),
   pixels_per_image INT,
   train_size INT,
   test_size INT
);

CREATE TABLE emotion_labels (
   emotion_id INT PRIMARY KEY,
   emotion_name VARCHAR(30)
);

CREATE TABLE video_processing_pipeline (
   step_id INT PRIMARY KEY,
   step_description TEXT
```

```
);

CREATE TABLE cnn_layers (
    layer_id INT PRIMARY KEY,
    layer_type VARCHAR(50),
    activation VARCHAR(30),
    description TEXT
);

CREATE TABLE data_augmentation (
    augmentation_id INT PRIMARY KEY,
    zoom_range DECIMAL(3,2),
    rotation_range INT,
    width_shift_range DECIMAL(3,2),
    height_shift_range DECIMAL(3,2),
    horizontal_flip BOOLEAN,
    vertical_flip BOOLEAN
);
```